

to repeat a task until it's done, is usually thought of as the opposite of recursion.

Whenever you find yourself asking, “Wouldn’t using a loop be easier?” the answer is almost certainly “Yes,” and you should avoid the recursive solution. Recursion can be tricky for both beginner and experienced programmers, and recursive code isn’t automatically “better” or “more elegant” than iterative code. Readable, easy-to-understand code is more important than any supposed elegance that recursion provides. However, on some occasions an algorithm cleanly maps to a recursive approach. Algorithms that involve tree-like data structures and require backtracking are especially suited for recursion. These ideas are further explored in Chapters 2 and 4.